

TERM-III | 2025-26

INTEGRATED GRIEVANCE REDRESSAL SYSTEM

Front End Development & UI Engineering

Team: K Keathana Vikashini & V Akshitha

| Executive Summary

A centralized digital platform designed to **empower citizens** by providing transparent channels for registering and tracking essential civic complaints.



Voice

Simple registration with multi-media evidence support.



Routing

Automated departmental forwarding for rapid resolution.



Trust

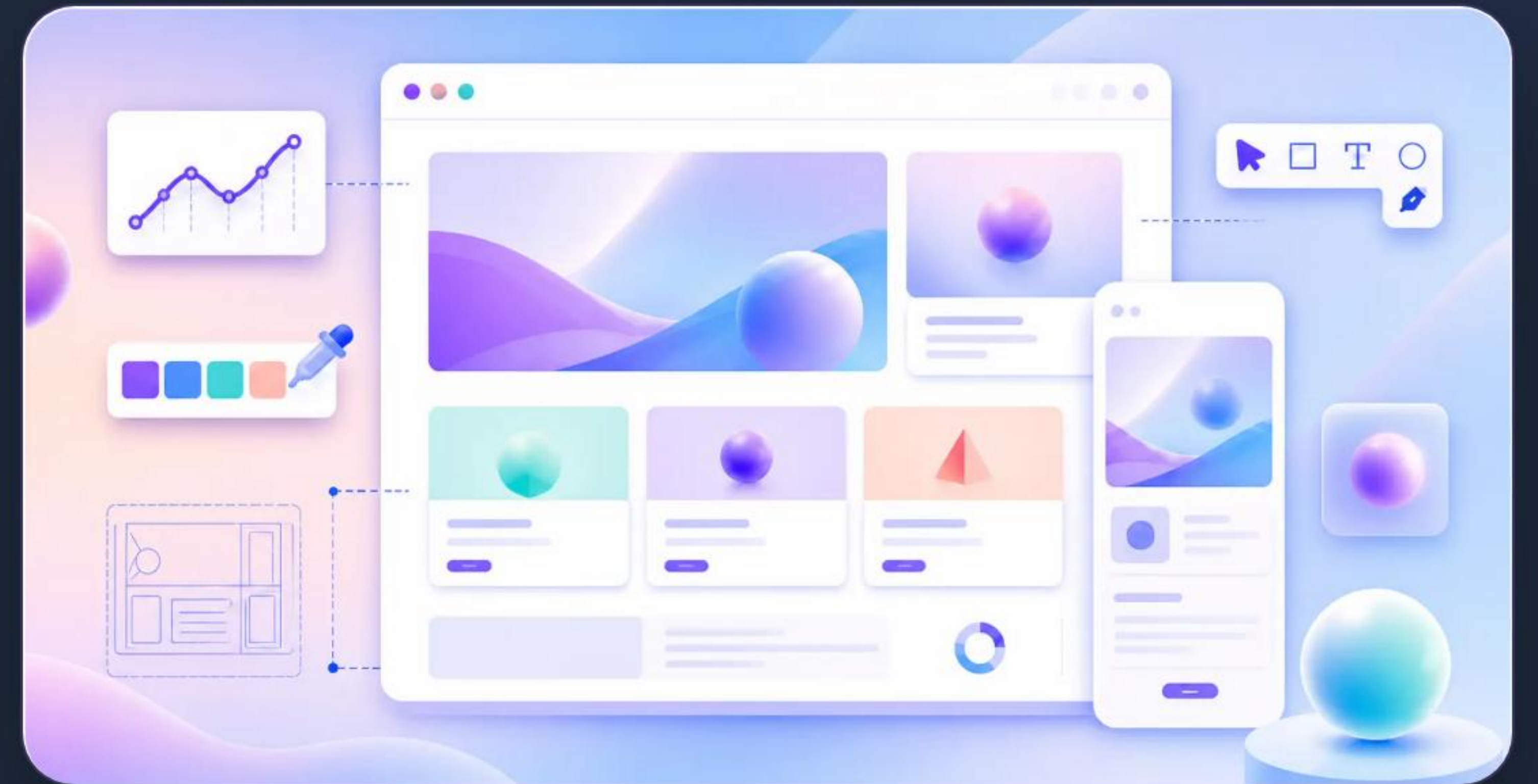
End-to-end status tracking with built-in SLA accountability.



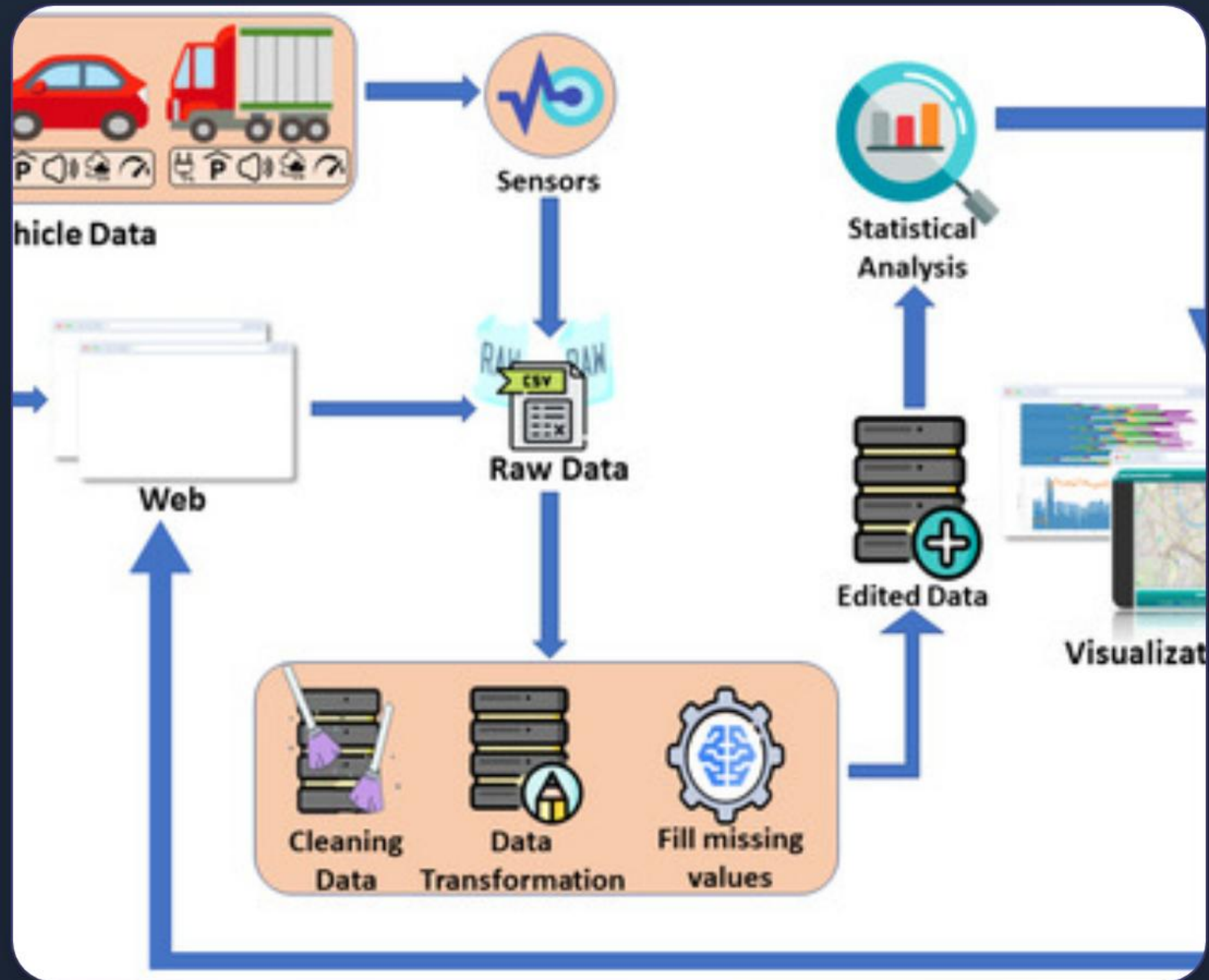
| The User-Centric Vision

Bridging the public-admin gap with modern **UI Engineering** patterns.

- ✓ High Transparency through live status steps.
- ✓ Minimalistic UI for diverse demographics.
- ✓ Mobile-first responsive architecture.



Scope of Public Services



Primary Target Sectors

- 💧 **Water Board:** Leakage and supply management.
- 🚧 **Infrastructure:** Potholes and drainage systems.
- ⚡ **Electricity:** Outage and streetlight tracking.

Architected for limitless departmental scalability.

Team Roles & Responsibilities

MODULE NAME	ASSIGNED MEMBER	CORE ENGINEERING TASKS
User-Side Module	V Akshitha	Smart Forms, Dept Routing, Priority Badging, SLA UI.
Admin-Side Module	K Keathana Vikashini	Admin Dashboard, Escalation Engine, Audit History, Reports.

Mentor: Dr. Shiva Krishna Nallabothu

Module 1: Citizen Interaction

-  **Dynamic Forms:** Precise data capture with field logic.
-  **Intelligent Routing:** Automated backend ticket distribution.
-  **Priority Matrix:** Color-coded urgency triaging.
-  **Status Stepper:** Real-time journey visualization.

UI Engineering Core

Implementing React Context API for reactive state management across submission flows, ensuring high-performance asynchronous data handling.

Module 2: Governance & Control

Admin Architecture

Designed for decision-makers to visualize systemic bottlenecks and optimize city-wide resource allocation in real-time.



Central Hub: Unified dashboard for complaint monitoring.



Escalation Flow: Time-triggered alerts for unresolved cases.



Audit Trail: Full history logging for transparency.

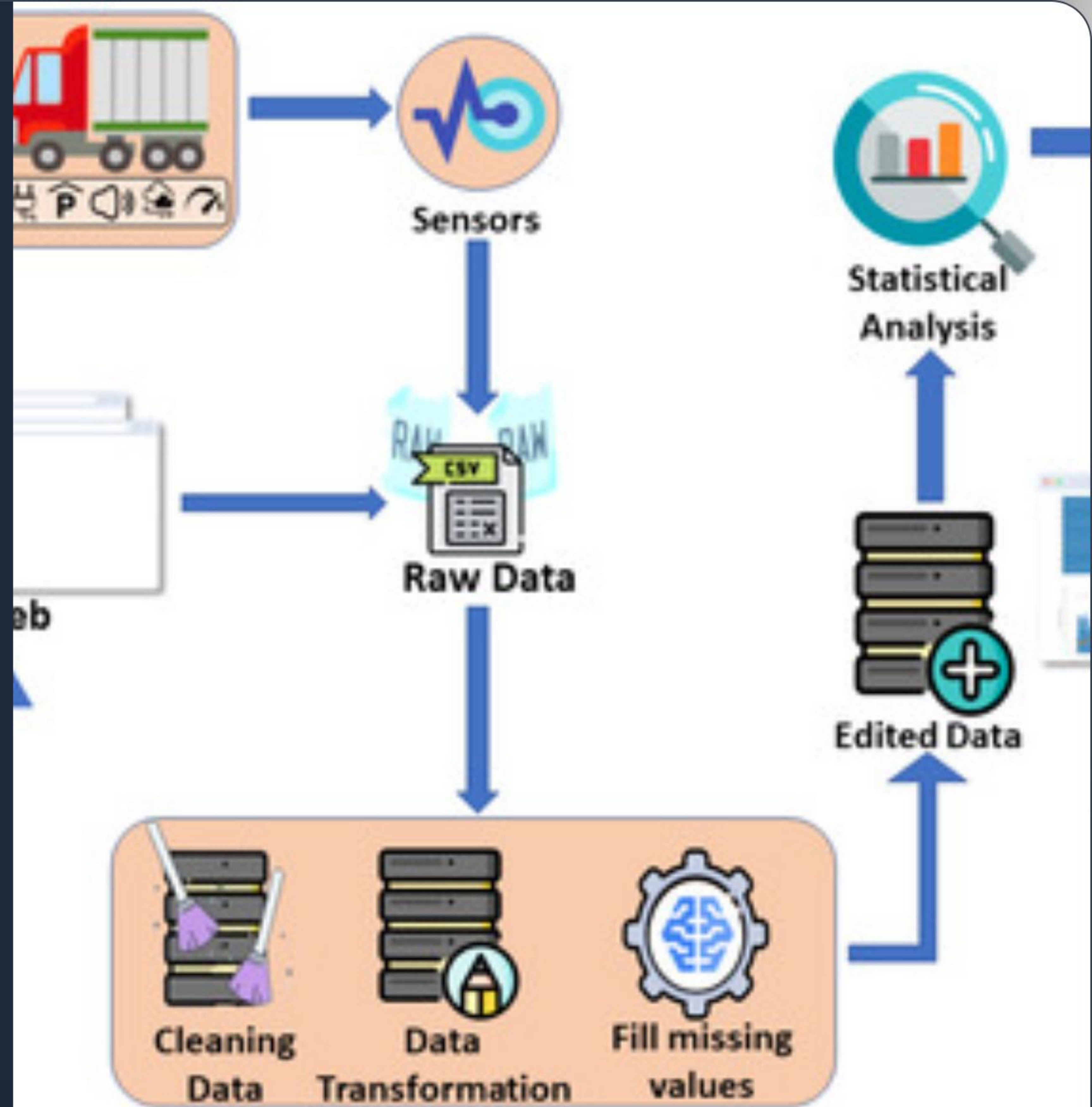


Reporting: One-click PDF/Excel data analytics.

Analytic Governance

Leveraging real-time telemetry to visualize trends.

The Admin Panel monitors resolution times and escalation triggers, identifying infrastructure hotspots to guide policy decisions.



| The Engineering Foundation



React



Node.js



MySQL

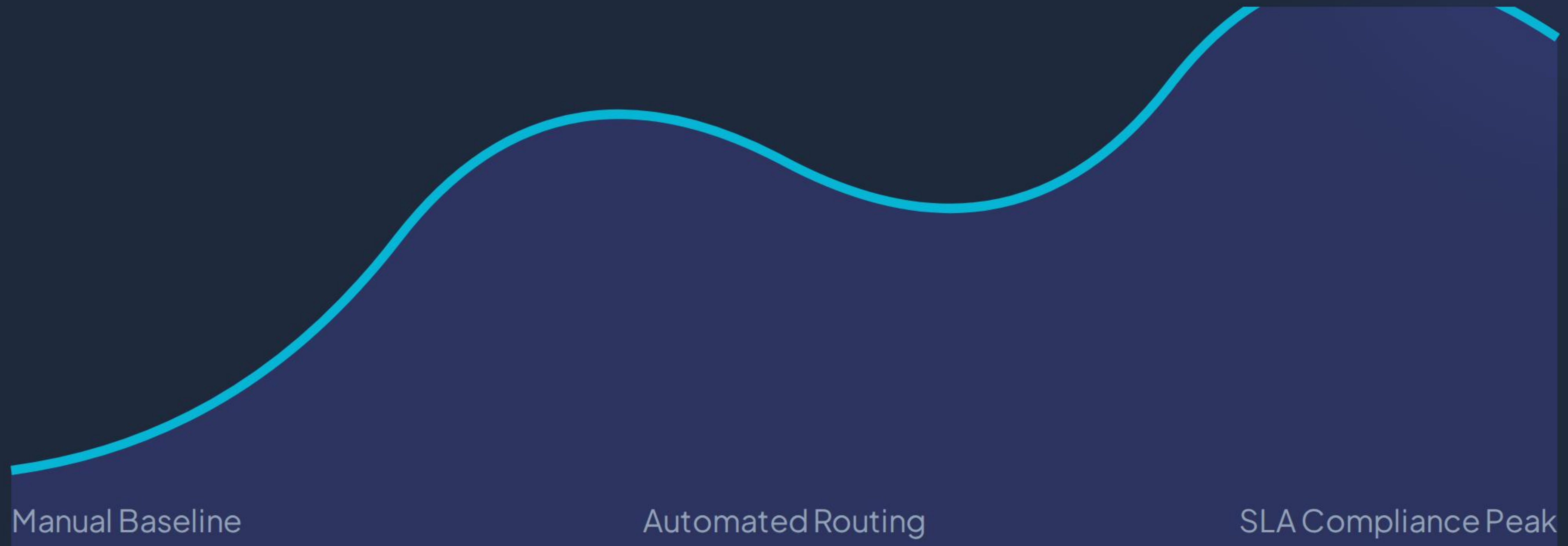


ES6+

Stack: MERN-influenced (MySQL) workflow with Axios for async operations and Redux/Context for state synchronization.

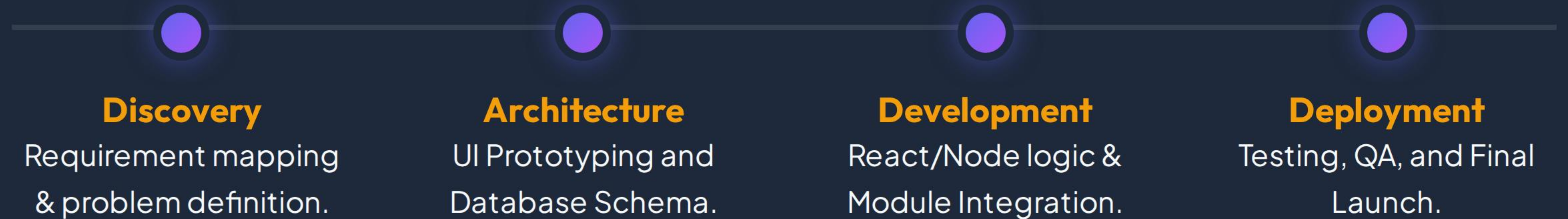
DevTools: NPM, Postman for API validation, and Tailwind/Bootstrap for responsive UI engineering.

SLA Compliance & Efficiency



Projected increase in resolution efficiency through automated escalation.

Implementation Roadmap



QUESTIONS?

INTEGRATED GRIEVANCE REDRESSAL SYSTEM

Keathana Vikashini

Team Lead

V Akshitha

Team Member



Image Sources



https://wpmanageninja.com/wp-content/uploads/2026/04/Web-Design-in-2026_-What-to-Anticipate-and-How-It-May-Look-1.webp

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